

Innovative designs from India

Texas Instruments (TI) has announced a scalable OMAP35x portfolio of devices available for the broad market that is ideal for manufacturers wanting to redefine the standard for advanced user interfaces, Web browsing, productivity and multimedia experience, making it easier than ever to access information and media.

The OMAP devices are an attractive solution for both big and small players. Target markets include portable consumer entertainment devices like portable media players and portable TV (broadcast and streaming TV content within the home), portable navigation devices, point-of-sale systems, patient monitoring and other medical applications, software-defined radio, IP-based communications, smart home media controllers and edutainment (eDictionaries).



Praveen K. Ganapathy,
director-DSP business
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Instruments, India



Ganesh Guruswamy,
vice president
and country
manager, Freescale
Semiconductor India

Freescale India design centre has completed many designs in last several years in the domains of wireless, networking and microcontrollers. MCF5225x MCU, Freescale's latest, was created by the India team in a record time. The MCU is well suited for a broad range of industrial networking, building/lighting control and medical applications that require high performance and connectivity options (Ethernet and USB).

Axis Aerospace and Technologies (AAT), formerly known as Jupiter Strategic Technologies, has developed a multi-radar data fusion system for the Southern Air Command of the Indian Air Force. AAT has designed and

developed the decision-making algorithms indigenously and the entire solution runs on a real-time operating system developed in-house. It claims to be the first and only company in India to integrate civil and military radars to generate a real-time air situation picture of the entire theatre of operations and provide decision support system for air defence.



Sharadhi Chandra
Babu P., vice president
(corporate), business
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Nirav Shah, director of
marketing, elfin chips

elfin chips' team of 50 engineers has worked on a design across two sites for a customer in USA. The design was of a transcoding appliance which houses multiple DSPs and a host processor running at higher speed to support video streaming to various display sizes at different bit rates through the Internet.

NXP has unveiled its cutting-edge technology solutions in the set-top box market. The STB 150 system solution, developed by NXP, delivers a complete production-ready platform for the fast growing digital cable TV services market. It integrates the powerful single-chip CX2448x MPEG-2 decoder and QAM demodulator for DVB-C reception with a TDA18252HN silicon tuner and complete software stack, giving cable operators a cost-effective silicon platform for a new range of advanced, feature-rich digital entertainment services. This is a low-cost reference design, designed for the Indian market keeping SD MPEG4 plus DVB-S2 focus. This hybrid solution also includes an Internet connection for IPTV solutions.



N.S. Murthy, director,
new business
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Semiconductors

DRAM, it is possible to scale eDRAM devices below 45 nm. As opposed to conventional eDRAMs which consist of a transistor and a capacitor, these capacitorless DRAMs employ one or two transistors, which can scale down to the 22nm process node.

According to Ganesh Guruswamy, vice president and country manager, Freescale Semiconductor, "One key trend in product design is going 'green,' i.e., lower power and longer battery life. Another trend is to develop single-chip solutions for any application by integrating analogue and digital technologies in one chip, i.e., analogue mixed-signal design."

Nirav Shah, director of marketing, elfin chips, says, "Video has taken a centre stage across all electronics, which is driving many architecture and design changes at chip and system level. We are witnessing 'real time' phenomena where everybody wants information/data 'now' at his fingertips."

What's ahead?

With Indian firms continuing to demonstrate skills in part life-cycle services such as design, verification and front-end/back-end services, Evolute's managing director Parag Mehta is hopeful that they will be able to bag increasing numbers of turnkey projects.

"Currently the captive units execute about two-thirds of all the work done, with the rest being done by third-party design houses. Over the next five years, the split between captive and third-party will become 50-50," predicts Kumar.

However, "to win turnkey designs, the Indian design companies will need to augment their analogue and mixed-signal capabilities. They will have difficulties doing this quickly if they tap only the Indian pool of engineers, and will need to look for international partnerships or acquisitions," says Partha Pratim Roy, advisor, Global Consumer Lab, Ericsson-Market Unit India & Sri Lanka. ●

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